Challenges and Opportunities for Survey Research in the Age of Generative AI: An Experience Report

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***Abstract*—Survey research, while common, has known challenges and limitations. In this paper, we discuss the challenges and concerns of conducting research using online survey in the age of widespread development and use of generative AI technologies. We base our discussion on our recent experiences conducting survey research to better understand software practitioners' knowledge of and experience with automated program repair (APR). In our efforts, we encountered both anticipated and unexpected challenges, much of which stemmed from advancements in AI-assisted automation (e.g., generative AI). For example, we found that many of the open ended responses provided were likely generated by AI. Based on our experiences, we outline how we mitigated or handled these challenges and discuss opportunities for improving survey design and administration to ensure high quality research and outcomes in the age of advanced and widely available AI technologies.**

***Index Terms*—human survey, online survey, generative AI, survey validation, automated program repair, AI-generated data, AI-generated responses**

I. Introduction

Collecting data directly from an audience of interest is an important part of research [1]. One of the most well-known options for collecting data is administering a survey. In practice, surveys are most often administered online given the flexibility and convenience in both collecting and analyzing the data [3], [7]. However, there are also documented challenges that come with using online surveys as a means for collecting data [3].

In our previous work, we conducted an online survey to gather insights from software practitioners on their knowledge of and experiences with automated program repair tools [11]. Our survey quickly received 800 responses, further suggesting the potential benefits of online surveys. However, despite our efforts in mitigating known challenges that come with administering online surveys, we lost over half of our responses due to concerns regarding data integrity.

In this paper, we will briefly summarize this previous survey, highlighting a few of the key insights that it helped reveal to us about software practitioners' automated program repair practices (Section II). Next, we share the strategies we used to broadly distribute our survey (Section III) and reflect on our high response rate (Section IV). We discuss the strategies we used to carefully validate responses (Section VII). We will also discuss how we used (or chose not to use) the potentially sensitive demographic data we collected (Sections VII & VIII). Additionally, we will describe how the use of generative AI may have impacted some participants' responses to our survey and share the strategies we used to detect and mitigate against this threat (Section VIII).

They key contributions of this paper are:

- an experience report on survey distribution and data analysis strategies in the era of generative AI,
- insights into the identification of AI-generated or -assisted survey responses, and
- implications for survey design and distribution of online data collection instruments.

II. Our Survey

We designed an online survey with 32 questions regarding experiences and requests for feedback on automated program repair (APR) tools from practitioners involved in software engineering tasks.[1]

To design our survey, we first searched well-known social media platforms that are frequently utilized by software practitioners, such as GitHub, Reddit, Hacker News, and Stack Overflow, to acquire initial insights into APR use in practice. These websites offered information on the perspectives of practitioners regarding automated program repair.

We gathered information on specific tools that practitioners are currently using as well as various blogs or posts (e.g., HackerNews posts[2]) to curate recent perspectives of practitioners about APR tools. Additionally, we examined prior studies that investigated software practitioners' attitudes towards APR before finalizing our survey questions and options for answers that we provided. We included a consent form in the beginning of our survey form which outlines all the terms and conditions related to participation in our study, providing respondents the

[1]This research is approved under IRBNet Number 2006552-1.
[2]https://news.ycombinator.com/item?id=19316228

choice to proceed by giving their consent. Our final survey included four sections.

1) Demographics: First, we collected participant *demographics*. The majority of the questions in this section were influenced by the 2022 StackOverflow Developer Survey.[3] This includes questions regarding the respondents' personal, social, and professional background as well as questions regarding their programming language use and preferences.

We also added questions regarding gender and race/ethnicity to investigate any potential associations between these demographics and APR tool usage or knowledge. Our goal was to engage a diverse sample of practitioners, so we included questions on physical and mental disabilities as well. To determine how representative our sample is of software professionals from various environments and with different experience levels, we added questions regarding the size of the organization, current job positions, and years of coding and professional software industry experience. We also added questions about the language that they have used over the past years for development or testing work and the language that they prefer working with for the next years. We provided options for a total of 10 languages including Java, Python, C++, C, SQL, Javascript. Lastly, we asked respondents about their experiences with APR tools within their teams and workplaces. For those who had experience using APR tools, we asked questions about specific tools and frequency of use.

2) Knowledge about APR: This section asked participants about their current knowledge of APR (e.g., awareness of existing tools) and use of APR tools within their team and workplace. We included a question to determine if they had experience using APR tools – those who did were asked a series of questions on their experiences.

3) Experience with APR: Only participants with prior APR tool usage experience saw this portion of the survey. Our goal was to determine how familiar or experienced our respondents are with using APR tools in practice. To elicit this information, we added a question regarding the type of tool(s) they typically use, providing the choice of any current tool(s) or any specific tool(s) developed for their team or workplace. If they chose the option(s) for specific tools developed for their team or workplace, we asked them to provide a brief description of the tool. Otherwise, we asked participants to provide the name of the existing APR tool that they use.

4) Opinions and Suggestions: The last portion of the survey asked respondents to provide their thoughts and recommendations on existing APR tools and techniques. This includes the pros and cons of current tools, alternatives they are using, and ideas for improvement. We also had questions regarding the frequency of bug fixing, sources for support, the impact various factors would have on their desire to use APR tools.

III. Survey Distribution Strategies

Previous studies suggest that triangulating efforts at dissemination across platforms can lead to moderately high response rates for surveys [2]. One common approach that we used for distributing our survey was to advertise on popular social media platforms X (formerly known as Twitter) and LinkedIn. We also made more targeted efforts at reaching practitioners by reaching out to personal contacts and collaborators that have direct connections with companies and practitioners. We were also intentional in contacting individuals who have vested interest in our research (e.g., interest in understanding software practice). To further encourage engagement, we offered respondents the opportunity to participate in a drawing for an Amazon gift card after completing the survey.

IV. Survey Responses

Our efforts yielded a total of 800 responses. Among those, 732 (92%) agreed to the conditions given in our consent form and filled out the remainder of the survey. We removed invalid responses and considered the remaining 331 valid responses for our analysis.

To contextualize our sample and explore correlations with APR awareness and experiences, as mentioned in Section II, we asked respondents a variety of background and demographic questions, including their age, gender and ethnicity. The majority of our respondents in our valid set of survey responses identified as male (59.21%) or female (40.48%), though we did have one valid respondent who identified as non-binary, genderqueer, gender non-conforming (0.30%). Respondents' ages ranged from 18 to over 50 years of age. We covered a wide range of respondents in terms of race/ethnicity including Hispanic or Latino/a, Indigenous Australian or Native American or Pacific Islander. We also saw diversity in the size of the respondents' places of work, ranging from freelancers or sole proprietors to companies with over 10,000 employees. Among our respondents, many reported having at least one physical or mental disability (43.50%). Respondents also reported different levels of overall and professional coding experiences ranging from a minimum of less than a year to a maximum of more than 50 years.

V. Anticipated Challenges

Although an anonymous online survey appeared to be the best option for getting adequate responses for our survey, prior work suggests it can in fact be hard to get high quality responses from online surveys [5]. Therefore, there are a number of challenges we anticipated in administering an online survey eliciting practitioner experiences.

A. Survey Distribution

The ability to get quality survey responses depends significantly on the distribution and advertising techniques. As we mentioned in Section III, we focused our distribution efforts on social media and personal contact associated with the software industry. The goal of advertising on social media was to reach a broad and larger audience, while the goal of engaging our personal contacts was to be more targeted and intentional about reaching our audience of interest.

[3] https://survey.stackoverflow.co/2022/

However, a number of challenges arose as we attempted to effectively disseminate our survey to our audience of interest. When advertising on social media, while it increases reach, it is less clear if and to what extent our advertising is reaching our audience of interest. Furthermore, as discussed in prior work, online surveys are highly susceptible to internet bots that are designed to seek out and complete surveys that offer compensation (as our did) [8]. So we also had concerns regarding if we are reaching real practitioners sharing their experiences or bots only aiming to potentially be compensated. In hindsight, we also realized additional challenges that are especially important for our future research plans. In particular, while we may be able to cast wide nets using online communities and more targeted efforts in advertising through contacts, achieving representation and diversity is still a challenge. It is also generally difficult to know what methods of advertising and distribution are the most and least successful.

B. Data Integrity

Another anticipated challenge we encountered was assessing or ensuring the integrity of survey responses, especially multiple-choice questions where there is little we can do on an individual or granular level to assess the quality of the response provided. Many of the data integrity-related challenges associated with collecting data via an online survey have been documented. For example, prior research suggests that in survey research it is not uncommon to have responses that were submitted hastily, thereby reducing confidence in the data provided [5]. Our survey was no exception.We received unusual responses for some demographic questions (e.g. selecting all options for race/ethnicity), as well as a few duplicate responses. We even had submissions where responses across questions were illogical (e.g., participants selecting ‘No’ for knowledge of APR but selecting ‘Yes’ for experience with APR). Sometimes, they also provided answers that did not map accurately with the question asked (e.g. responding ‘email’ when asked about APR tools they have used). Filtering these invalid responses out was an anticipated challenge for us; we discuss how we attempted to mitigate these challenges in Section VII.

C. Collecting Sensitive Data

One common concern for human surveys is how to collect and use respondents’ sensitive and/or personal information. Even if the whole survey is anonymous, if we want to add a compensation for participants, then we need to get their contact information. This is a very general issue of privacy for surveys. Therefore, it was our anticipated challenge to ensure we collect, store and use participant’s personal information with care.

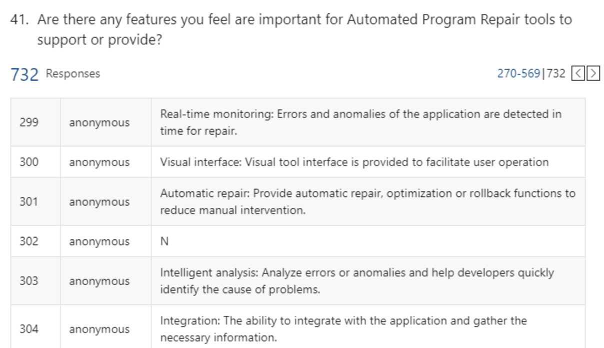
VI. Unanticipated Challenges

Besides the common anticipated challenges we encountered for our survey responses, we encountered a few unexpected challenges while trying to filter out invalid responses.

A. AI-generated Responses

In our efforts, we uncovered a challenge not yet documented in prior work that can impact data analysis and outcomes, which was identifying human- vs. AI-generated open ended responses. More specifically, in organizing our survey data, we realized that respondents may have been using AI assistants (most likely ChatGPT) to aid in completing the survey.

This concern arose when we noticed a common pattern emerging across responses. Several respondents used a very similar and particular format, *<Keyword> : <Definition>*. For instance, when asked “Are there any features you feel are important for Automated Program Repair tools to support or provide” one responded wrote *‘Multi-language support: Support for applications in multiple programming languages, such as Java, Python, JavaScript, etc.’*. Figure 1 shows other similar responses.


41. Are there any features you feel are important for Automated Program Repair tools to support or provide?

732 Responses 270-569|732

299	anonymous	Real-time monitoring: Errors and anomalies of the application are detected in time for repair.
300	anonymous	Visual interface: Visual tool interface is provided to facilitate user operation
301	anonymous	Automatic repair: Provide automatic repair, optimization or rollback functions to reduce manual intervention.
302	anonymous	N
303	anonymous	Intelligent analysis: Analyze errors or anomalies and help developers quickly identify the cause of problems.
304	anonymous	Integration: The ability to integrate with the application and gather the necessary information.

Fig. 1. Open-ended question responses from participants suspected to be AI-generated

This was a new and unique concern for us that we did not anticipate. Therefore, we needed to find a way to confirm our suspicion for usage of AI-assisted tool and then discard those submissions from analysis.

B. Using Sensitive Data

We collected demographics data from our participants to contextualize our sample and determine whether there are correlations or relationships between demographics and their knowledge of or experience with APR. We tried to focus our demographic and background questions on information we felt would be relevant to understanding APR knowledge or experience, while also considering the value in understanding how representative our findings are the diverse population of potential APR users. We did not find any significant relationships between demographic information, but reported on them in the first draft of our publication. One of the reviewers commented on this, specifically stating uncertainty around collecting demographic information that did not explicitly relate to the goals or results. This prompted reflection from us on survey design and clarity around the rationale behind the information solicited.

VII. Mitigating Anticipated Challenges

While it is impossible to ensure the removal of all invalid or low quality responses, we took the following steps in our attempts to mitigate the threats to validity that come with administering online surveys.

A. Survey Distribution

To disseminate help mitigate the issue of reaching our audience of interest, which was software practitioners, we used hash tags (e.g., `#SoftwareEngineering`) as a mechanism to increase the likelihood that our post would organically reach like-minded individuals [12]. To further increase the our odds of engaging software practitioners, we only posted the survey on our professional social media accounts where we have connections with professionals. We avoided Facebook as everyone on the research team uses Facebook for personal reasons. Given the challenges inherent to administering a survey on social media, we also connected with our personal contacts in industry to share our survey.

B. Data Integrity

We took a number of measures in an attempt to mitigate data integrity issues in our final set of survey responses. As we mentioned in Section V-B, one key challenge we anticipated was dealing with potentially inattentive respondents or responses done hastily. The first effort we made was in the design of our survey. Inspired by a prior work we encountered in our efforts [9], we added an **attention question** to ensure we include responses where respondents gave the survey due consideration. As shown in Figure 2, in our attention question, we asked respondents to select "Occasionally" among the options provided. This alone removed 289 potentially invalid or low quality survey submissions.

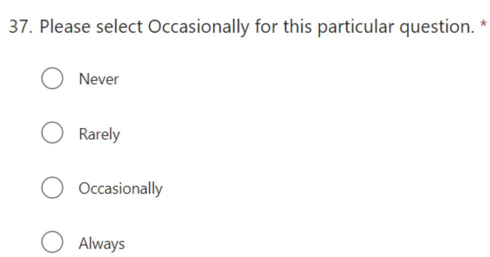


Fig. 2. **Attention question added in the survey**

Along with the attention question, we also used **survey response time** to ensure the quality of our data. Prior work suggests that respondents who take less than 10 seconds per question may not be giving due consideration to provide useful or accurate responses [9]. Given our survey was approximately 32 questions (26 questions for practitioners with no prior experience in using any APR tool), this suggested a threshold of approximately 4 minutes for quality responses. However, before distributing our survey, we piloted with students in our lab and found that quality responses can be achieved when response time is over 3 minutes. Therefore, we set our threshold for quality responses at 3 minutes or more; this led to the removal of an additional 125 survey responses.

As another form of screening, we also made sure to pay close attention to **excessive response selection** in multiple choice questions that allowed for the selection of more than one option. This was mostly this case for our demographic questions where we wanted to allow for respondents to describe their identity in as much detail as they deemed necessary. We generally did not find this to be an obvious or previously cited challenge, however, we did find that checking these kinds of questions can also help mitigate potential data integrity issues.

For example, we asked respondents to provide their race and ethnicity where we provided a list of 22 options, excluding none and their preference of not mentioning. While most provided what would be consider realistic and believable responses, we had one respondent who selected the combination of "Hispanic or Latino/a, Asian, Indian, European, White, Middle Eastern, North American, South American, African, Southeast Asian, South Asian, Multiracial, Black, East Asian, Biracial, Caribbean, North African, Central American, Central Asian, Ethnoreligious group, Indigenous (such as Native American or Indigenous Australian), and Pacific Islander". We also had a questions regarding physical and mental disabilities where we found that two respondents selected 7+ options at once. Without the ability to adequately confirm or validate the responses or where they came from, we decided to remove these responses as well. These efforts resulted in our removing 3 survey submissions.

To mitigate any additional potential for data integrity issues, we also excluded responses if they:

- declined to give their consent to fill out the survey form [68 responses],
- provided responses that did not clearly or accurately map to the question (e.g., answering "email" as an APR tool they have used) [72 responses],
- provided duplicate responses [7 responses],
- selected inconsistent answers, which is 'No' for knowledge of APR tools but 'Yes' to experience using APR tools [11 responses]

C. Collecting Sensitive Data

To mitigate challenges with collecting sensitive data, we first made sure to design our survey such that the collection of personally identifiable information (PII) is separate from the other questions, including demographic, that we asked. The only PII we collected was respondents' email addresses for compensation; no additional information was collected that can allow someone to be directly identified and it was not required for respondents to provide their email. We clearly elaborated in our consent form that we would be storing all data on secure servers and remove all PII once compensation is completed. All aspects of our study were approved by the Institutional Review Board (IRB) approval at our university.

VIII. Handling Unanticipated Challenges

While our efforts at mitigating expected challenges helped increase confidence in our data collection efforts, we still encountered challenges and concerns we did not explicitly prepare for. More specifically, we encountered issues of AI-generated responses and concerns regarding use of sensitive data in reporting.

A. AI-generated Responses

As discussed in Section VI, a challenge the emerged in our data analysis efforts was identifying and properly handling survey submissions that included responses potentially generated by an AI assistant, like ChatGPT. We noticed several participants of our survey had similar patterns for their open-ended responses. This pattern was familiar to one of the researchers on our research team and, for them, pointed to possibly using an AI assistant like ChatGPT to generate a response. Therefore, we attempted to use ChatGPT to generate answers for our open-ended questions and found similar patterns among the answers (e.g., *<Keyword> : <Definition>*).

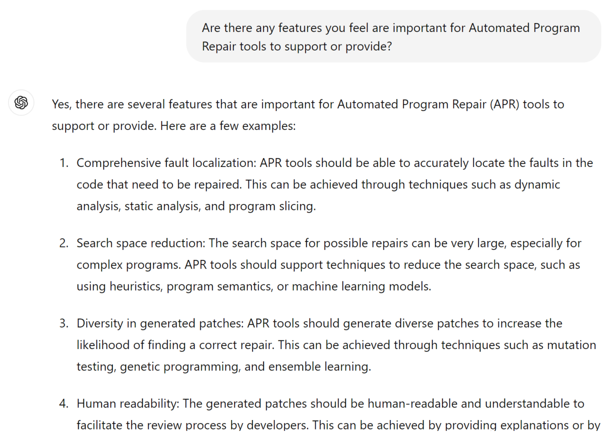


Fig. 3. Open-ended question responses directly from ChatGPT

Figure 3 depicts the similarly formatted responses that ChatGPT provided. Having never encountered this kind of issue before, we were unsure of whether we should remove only the open-ended responses on these submissions or remove the entire submission. We were concerned that we may miss out on relevant and useful information be removing the entire submissions, however, we also had concerns regarding the due diligence respondents who were using AI assistants may be giving to providing responses. However, not knowing the extent to which respondents used AI to complete the survey, and unsure of how to properly handle this kind of issue, we removed any survey submissions with what we suspected to include AI-generated responses. Along with the additional effort in analyzing and determining data quality, this led to the removal of 5 submissions.

B. Using Sensitive Data

We collected responses to a standard set of demographic questions following our survey to better understand whether our sample represents the general population of the software development community. This detailed information is presented in this paper (Section IV) solely to contextualize our sample. Analyzing the information of disability, gender, and race were not the focus of our previous study, and therefore we did not present any hypothesis or tests based on these demographics. However, this lead to comments from reviewers about the necessity of collecting these information if not used. So we removed reporting of various aspects of our sample. However, we feel these information are critical to collect as it may enable us to conduct future, more focused studies exploring. For example, we found that a mentionable subset of our sample reported having physical and/or mental disabilities (43.50%). This provides motivation and insight for future research on how APR tools or large language models (LLMs) use among practitioners with disabilities or mental health issues.

IX. Lessons Learned & Discussion

Despite the challenges that come with administering online surveys, the potential benefits and importance of this kind of data collection are undeniable. Our efforts bring to light novel considerations and implications for how we design, administer, and analyze online surveys in the age of ubiquitous AI.

A. Identifying & Handling AI Generated Responses

The goal of our survey research was to better understand the experiences of *people* who use APR tools. With the emergence of technologies like ChatGPT that leverage large language models (LLMs) for generating text, it has become even easier to use automation to support the completion of online surveys, thus making it more difficult for the researcher to determine or validate the quality of the overall submission. In our efforts, we encountered unexpected challenges in identifying and handling AI-generated responses. This was a necessary validation step, especially considering the potential for hallucinations which may lead us to have factually incorrect, fabricated, or misleading information considered in our analysis. However, had we not noticed the pattern in our open ended responses we may not have done as much due diligence in this regard.

The availability and widespread use of AI technologies like ChatGPT increase the concerns around administering online surveys, especially those that offer any form of compensation [8]. Mitigating a challenge of this nature requires considerations in survey design, dissemination, and validation. For example, when designing our surveys it may be necessary to explicitly discourage the use of AI assistant in completing the survey. For surveys that provide compensation, we can add that responses that appear to have been generated by AI will not be considered and therefore do not qualify for compensation. This also increases the importance of seeking collaborations and partnerships that can facilitate direct engagement with audiences of interest, which heightens that chances of the survey getting valid responses.

B. Reducing Data Integrity Challenges

One of the major challenges with online surveys is mitigating, identifying, and appropriately handling issues with data integrity. We discussed a few mechanisms we employed in this direction; however, despite our efforts we still found numerous issues (both expected and unexpected) that led to the reduction of our overall data sample.

There exists platforms aimed at supporting survey research through recruitment of respondents, such as Qualtrics [4] and Prolific [5]. While this is encouraging, these are general purpose platforms that may not be well-suited for studies like ours that are targeting individuals with a specific occupation. Furthermore, prior efforts that have made use of these platforms suggest similar challenges with validating the integrity of data collected [13], [14]. In fact, one study explicitly compared data quality from CS students recruited from a mailing list and respondents recruited from Prolific and found that CS students outperformed Prolific respondents in the programming tasks provided [14].

While using students is a common practice in academic research, it is also one that is often criticized for not reporting on experiences relevant to practice. This is especially the case for studies like ours where the goal is to understand how practitioners are building real world systems. One way to handle this kind of issue is to, as a community, be more intentional about academia-industry synergy and collaborations that can facilitate the recruitment of practitioners for studies on practice. To this end, prior efforts have explored ways we can build connections between academics who do practical research and practitioners [10]. Unfortunately, they experienced little success, raising questions regarding what it looks like to effectively facilitate the collaboration of academia and industry in research and development.

As we work to determine the best ways to engage our audiences of interest in the collection of survey data, one intermediate solution that can support researchers is to increase our awareness regarding methods that are most and least effective. One concrete option for survey design is to consider including mechanisms such as tracking questions (e.g., "where did you hear about this survey?") that provide insights into how those who provide high quality responses are finding the survey. This can be especially useful when conducting research that relies on diversity in the responses.

C. The Value in Collecting Sensitive Data

The landscape of software practitioners is vast and encompasses a spectrum of identities. Even when a study is not focused on a specific demographic or community, it is important we understand the representation in our survey data and use the opportunities when presented to discuss implications for subgroups in our audience of interest. Furthermore, prior work has found that respondents are more likely to provide demographic information when its integrated into the main study and no less likely to participate when they are present [15]. Collecting insights on demographics present (and not present) in our human-centric efforts can amplify gaps in our understanding and motivate targeted research efforts. For example, our research efforts found that many software practitioners are using ChatGPT to support their program repair efforts. We also had a significant portion of our sample that noted having physical and/or mental disabilities. This has motivated our current and future efforts aimed at understanding the impact of AI assistant use on practitioner well-being [11].

[4] https://www.qualtrics.com/research-services/online-sample/

[5] https://www.prolific.com/

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